

## Lecture 1

### Introduction to Probabilistic Model

Consider a random vector  $X = (X_1, X_2, \dots, X_d)$

Want to model the relationship  
between these variables.

Probabilistic Models: Relate by joint distribution  
Generative  $p(x) = p(x_1, \dots, x_d)$

#### Our Objective

Suppose there is a true  $p^*$ .  
we approximate it by  $p \in \mathcal{P}$  s.t.  $p \approx p^*$  Our model

好处, 当我们有了  $p(x, y)$ ,  $p(x, c)$ ,

我们可以 regression  $p(y|x) = \frac{p(x, y)}{\int p(x, y) dy}$

classification  $p(c|x) = \frac{p(x, c)}{\sum_c p(x, c)}$

#### 模型的假设

1. We always assume that samples  $x^{(i)}$  are iid.
2. "Parametrized Distribution"

$$\mathcal{P} = \{p(x|\theta) : \theta \in \Theta\}$$

## Likelihood Function

$$D = \{x^{(1)}, \dots, x^{(n)}\}$$

Likelihood  
Function

$$L(\theta; D) = p(D|\theta) = \prod_{i=1}^n p(x^{(i)}|\theta)$$

Log-Likelihood  
Function

$$\ell(\theta; D) = (\log L(\theta; D)) = \sum_{i=1}^n \log p(x^{(i)}|\theta)$$

One way of finding estimator — MLE

$$\hat{\theta}_{MLE} = \underset{\theta \in \Theta}{\operatorname{argmax}} L(\theta; D) = \underset{\theta \in \Theta}{\operatorname{argmax}} \ell(\theta; D)$$

e.g.  $x^{(1)}, \dots, x^{(n)} \sim \text{Bern}(\theta)$

$$\begin{aligned} \ell(\theta; D) &= (\log L(\theta; D)) = \log \prod_{i=1}^N \theta^{x^{(i)}} (1-\theta)^{(1-x^{(i)})} \\ &= \sum_{i=1}^N x^{(i)} \log \theta + (1-x^{(i)}) \log(1-\theta) \\ &= \sum_{i=1}^N x^{(i)} \log \theta + \left(N - \sum_{i=1}^N x^{(i)}\right) \log(1-\theta) \end{aligned}$$

$$\frac{\partial \ell(\theta; D)}{\partial \theta} = 0 \Rightarrow \frac{\sum_{i=1}^N x^{(i)}}{\theta} = \frac{N - \sum_{i=1}^N x^{(i)}}{1-\theta} \Rightarrow \hat{\theta}_{MLE} = \frac{\sum_{i=1}^N x^{(i)}}{N}$$

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Sufficient Statistic

Defn.

$T(x)$  is sufficient  $\Leftrightarrow p_{\theta}(x | T(x)=t)$

Independent of  $\theta$

感性: The statistic that conveys the exact same information about parameter as the whole data.

Thm.

## Fisher - Neyman Factorization Theorem

$T(x)$  is sufficient  $\iff p(x|\theta) = h(x) g_\theta(T(x))$   
for some function  $h(\cdot)$  (depend on  $\theta$ )  
 $g_\theta$

在 sufficient statistic 上有  
性质的 pdf/pmf 家族

## Exponential Families

Defn.

Density of a member of exponential families is of the form

$$p(x|\eta) = h(x) \exp \{ \eta^T T(x) - A(\eta) \}$$

$T(x)$ : sufficient statistic

$\eta$ : natural parameter

$A(\eta)$ : log-partition function

$h(x)$ : measure  
carrying

e.g.

$$p(x|\theta) = \theta^x (1-\theta)^{1-x}$$

$$X \sim \text{Bern}(\theta)$$

$$= \exp \{ \log [\theta^x (1-\theta)^{1-x}] \}$$

$$= \exp \{ x \log \theta + (1-x) \log (1-\theta) \}$$

$$= \exp \{ x \log \theta - x \log (1-\theta) + \log (1-\theta) \}$$

$$= \exp \{ x \log \frac{\theta}{1-\theta} + \log (1-\theta) \}$$

$$\eta = \log \frac{\theta}{1-\theta}$$

$$A(\eta) = \log (1 + e^\eta) \quad (A(\eta) = -[\log (1-\theta)])$$

$$h(x) = 1$$

$$T(x) = x$$

Tip 怎么解  $A(\eta)$

$$A(\eta) = -\log(1-\theta) \quad \eta = \log\left(\frac{\theta}{1-\theta}\right)$$

① 先把  $\theta$  变成  $\eta$

$$e^\eta = \frac{\theta}{1-\theta} \Rightarrow e^\eta - e^\eta \theta = \theta \\ \Rightarrow e^\eta = (1+e^\eta)\theta \Rightarrow \theta = \frac{e^\eta}{1+e^\eta}$$

② 代入  $A(\eta) = -\log(1-\theta)$

$$A(\eta) = -\log\left(1 - \frac{e^\eta}{1+e^\eta}\right) = -\log\left(\frac{1}{1+e^\eta}\right) = \log(1+e^\eta)$$

结果

发现 ①  $A'(\eta) = \frac{e^\eta}{1+e^\eta} = \theta$

$$\text{and } E[T(X)] = E[X] = \theta$$

$$\textcircled{2} A''(\eta) = \frac{e^\eta}{1+e^\eta} \left(1 - \frac{e^\eta}{1+e^\eta}\right) = \theta(1-\theta)$$

$$\text{Var}[T(X)] = E[(X - E[X])^2] = \theta(1-\theta)$$

$\uparrow$   
 $X$

why?

Thm

$$A'(\eta) = E[T(X)] \quad \text{when } X \sim p(x|\eta) = h(x) \exp\{\eta^T T(x) - A(\eta)\}$$

Pf:

$$E[T(x)] - A'(\eta) = \int T(x) p(x|\eta) dx - A'(\eta)$$

$$= \int T(x) p(x|\eta) dx - A'(\eta) \int p(x|\eta) dx$$

$$= \int (T(x) - A'(\eta)) p(x|\eta) dx$$

$$= \int (T(x) - A'(\eta)) h(x) \exp\{\eta^T T(x) - A(\eta)\} dx$$

$$= \int \frac{\partial}{\partial \eta} p(x|\eta) dx = \frac{\partial}{\partial \eta} \int p(x|\eta) dx = \frac{\partial}{\partial \eta} (1) = 0$$

$$\therefore E[T(x)] = A'(\eta)$$

Th  $A''(\eta) = \text{Var}(T(x)) \quad x \sim p(x|\eta)$

Pf:  $A'(\eta) = E[T(x)]$

$$\begin{aligned} A''(\eta) &= \frac{\partial}{\partial \eta} E[T(x)] = \frac{\partial}{\partial \eta} \int T(x) p(x|\eta) dx \\ &= \int T(x) \frac{\partial}{\partial \eta} p(x|\eta) dx \\ &= \int T(x) [T(x) - A'(\eta)] p(x|\eta) dx \\ &= \int T(x)^2 dx - \left[ \int T(x) p(x|\eta) dx \right] [A'(\eta)] \\ &= E[T(x)^2] - (E[T(x)])^2 \end{aligned}$$

Exponential Family as MLE Estimator?

Recall:  $p(x|\eta) = h(x) \exp\{\eta^T T(x) - A(\eta)\}$   $x^{(1)}, \dots, x^{(n)} \text{ iid } p(x|\eta)$

$$\begin{aligned} \ell(\eta; D) &= \log L(\eta; D) = \sum_{i=1}^n \log p(x^{(i)}|\eta) \\ &= \sum_{i=1}^n \log [h(x^{(i)}) \exp\{\eta^T T(x^{(i)}) - A(\eta)\}] \\ &= \sum_{i=1}^n (\log h(x^{(i)}) + \eta^T T(x^{(i)}) - A(\eta)) \\ &= \sum_{i=1}^n \log h(x^{(i)}) + \eta^T \sum_{i=1}^n T(x^{(i)}) - n A(\eta) \end{aligned}$$

$\frac{\partial \ell(\eta; D)}{\partial \eta}$  or  $\nabla_{\eta} \ell(\eta; D)$  in higher dimension,

$$= 0 + \sum_{i=1}^n T(x^{(i)}) - n A'(\eta) = 0$$

$$\Rightarrow A'(\hat{\eta}_{MLE}) = \frac{1}{n} \sum_{i=1}^n T(x^{(i)})$$

Tip: We also know that

$$A'(\eta) = E[T(x)]$$

$\therefore \hat{\eta}_{MLE}$  satisfies  $E[T(x)] = \frac{1}{n} \sum_{i=1}^n T(x^{(i)})$

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## MLE and Exponential Families

### N-sample example: Multinomial Distribution

A random variable  $X \sim \text{Multinomial}(q)$  where  $q \in \mathbb{R}^k$

$X \in \{1, \dots, k\}$  and  $P(X=i) = q_i$

$q_i \geq 0 \quad \sum q_i = 1$

动机

We want to write the whole dataset as an exponential family.

$$p(x|\eta) = h(x) \exp\{\eta^T T(x) - A(\eta)\}$$

$$p(D|\eta) = h(D) \exp\{\eta^T T(D) - A_D(\eta)\}$$

① First way, ~~per D~~  
(不管自由度)

$$p(D, q) = \prod_{i=1}^N p(x^{(i)}|q) = \prod_{i=1}^N q_1^{\mathbb{1}\{x^{(i)}=1\}} \dots q_k^{\mathbb{1}\{x^{(i)}=k\}}$$

$$= \prod_{i=1}^N \prod_{j=1}^k q_j^{\mathbb{1}\{x^{(i)}=j\}}$$

$$= \exp\left(\log \left( \prod_{i=1}^N \prod_{j=1}^k q_j^{\mathbb{1}\{x^{(i)}=j\}} \right)\right)$$

$$= \exp\left\{ \sum_{i=1}^N \sum_{j=1}^k \mathbb{1}\{x^{(i)}=j\} \log q_j \right\}$$

$$\text{Let } N_i = \sum_{i=1}^N \mathbb{1}\{x^{(i)}=i\} \quad = \exp\left\{ \sum_{j=1}^k \sum_{i=1}^N \mathbb{1}\{x^{(i)}=j\} \log q_j \right\}$$

$$= \exp\left\{ \sum_{j=1}^k \log q_j \sum_{i=1}^N \mathbb{1}\{x^{(i)}=j\} \right\}$$

$$= \exp\left\{ \sum_{j=1}^k \log q_j N_j \right\}$$

then,  $h(D) = 1$       $\eta = \begin{pmatrix} \log q_1 \\ \vdots \\ \log q_k \end{pmatrix}$

$$T(D) = \begin{pmatrix} N_1 \\ \vdots \\ N_k \end{pmatrix}$$

问题是:  $A_D(\eta) = 0$

没有  $A_D(\eta)$  了.

原因:  $\sum q_i = 1 \rightarrow$  只有  $k-1$  自由度

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② Second way (修正好带上自由度)

竟然  $\sum q_k = 1$  . note  $q_k = 1 - \sum_{i=1}^{k-1} q_i$

$$\begin{aligned}
 p(D; q) &= \exp \left\{ \sum_{i=1}^K \log q_i N_i \right\} \\
 &= \exp \left\{ \sum_{i=1}^{K-1} \log q_i N_i + N_K \log q_K \right\} \quad \leftarrow N_K = N - \sum_{i=1}^{K-1} N_i \\
 &= \exp \left\{ \sum_{i=1}^{K-1} \log q_i N_i + \left( N - \sum_{i=1}^{K-1} N_i \right) \log q_K \right\} \\
 &= \exp \left\{ \sum_{i=1}^{K-1} N_i \log \frac{q_i}{q_K} + N \log q_K \right\} \quad \leftarrow q_K = 1 - \sum_{i=1}^{K-1} q_i \\
 &= \exp \left\{ \sum_{i=1}^{K-1} N_i \log \frac{q_i}{1 - \sum_{j=1}^{K-1} q_j} + N \log \left( 1 - \sum_{j=1}^{K-1} q_j \right) \right\}
 \end{aligned}$$

then  $T(D) = \begin{pmatrix} N_1 \\ \vdots \\ N_{K-1} \end{pmatrix}$   $\eta = \begin{pmatrix} \log \frac{q_1}{1 - \sum_{j=1}^{K-1} q_j} \\ \vdots \\ \log \frac{q_{K-1}}{1 - \sum_{j=1}^{K-1} q_j} \end{pmatrix}$   $h(D) = 1$

Transform  $A(\eta)$   
 $\eta_i = \log \frac{q_i}{1 - \sum_{j=1}^{K-1} q_j} \Rightarrow e^{\eta_i} = \frac{q_i}{1 - \sum_{j=1}^{K-1} q_j} = \frac{q_i}{q_K}$

$$\sum_{i=1}^K q_i = 1 \Rightarrow \sum_{i=1}^K \frac{q_i}{q_K} = 1/q_K \Rightarrow \cancel{\frac{1}{q_K}} + \sum_{i=1}^{K-1} \frac{q_i}{q_K} = \frac{1}{q_K}$$

$$\Rightarrow \frac{1}{q_K} = 1 + \sum_{i=1}^{K-1} \frac{q_i}{q_K} = 1 + \sum_{i=1}^{K-1} e^{\eta_i}$$

$$A(\eta) = -N \log \left( 1 - \sum_{i=1}^{K-1} q_i \right)$$

$$= -N \log q_K = N \log \frac{1}{q_K} = N \log \left( 1 + \sum_{i=1}^{K-1} e^{\eta_i} \right)$$

Whole data set  $\longleftrightarrow p(D|\eta) = \exp \{ \eta^T T(D) - A(\eta) \}$

$$\eta_i = \log \frac{q_i}{1 - \sum_{j=1}^{K-1} q_j} \quad T(D)_i = N_i \quad A(\eta) = N \log \left( 1 + \sum_{i=1}^{K-1} e^{\eta_i} \right)$$

By the formula,  $\nabla_{\eta} A(\eta)_{MLE} = T(D)$

then for one  $i$ ,

$$\frac{\partial A(\eta)}{\partial \eta_i} = N \frac{e^{\eta_i}}{1 + \sum_{j=1}^{K-1} e^{\eta_j}} = N \frac{\frac{q_i}{q_K}}{\frac{1}{q_K}} = N q_i$$

$$\therefore N \hat{q}_i = N_i$$

$$\hat{q}_{i, MLE} = \frac{N_i}{N} \quad \forall i = 1, \dots, K-1$$

Tip

为什么  $\nabla_{\eta} A(\hat{\eta}_{MLE}) = T(D)$  对于 whole data

Pf: 对于 whole data 而且,

$$P(D|\eta) = h(D) \exp\{\eta^T T(D) - A_D(\eta)\}$$

$$\text{then, } \nabla_{\eta} \ell(\eta; D) = \nabla_{\eta} \log P(D|\eta)$$

$$= \nabla_{\eta} [\log h(D) + \eta^T T(D) - A_D(\eta)]$$

$$= T(D) - \nabla_{\eta} A_D(\eta) = 0$$

$$\nabla_{\eta} A_D(\eta)$$

$$\therefore \nabla_{\eta} A_D(\hat{\eta}_{MLE}) = T(D)$$

③ Third way, 我们直接用普通MLE.

$$P(D; q) = \exp\left\{\sum_{i=1}^k N_i \log q_i\right\}$$

$$\Rightarrow \ell(q; D) = \sum_{i=1}^k N_i \log q_i$$

我们的目标  $\max_q \ell(q; D)$  with constraint  $\sum_{i=1}^k q_i = 1$

$$\text{i.e. } g(q) = 1 - \sum_{i=1}^k q_i = 0$$

We write Lagrange Multiplier Theorem Equations.

$$\mathcal{L}(q; D) = \ell(q; D) + \lambda g(q)$$

$$= \sum_{i=1}^k N_i \log q_i + \lambda (1 - \sum_{i=1}^k q_i)$$

$$\frac{\partial \mathcal{L}(q; D)}{\partial q_i} = \frac{N_i}{q_i} + (-\lambda) \Rightarrow \lambda = \frac{N_i}{q_i} \Rightarrow q_i = \frac{N_i}{\lambda}$$

and we know the constraint  $1 - \sum_{i=1}^k q_i = 0$

$$\Rightarrow 1 - \sum_{i=1}^k \frac{N_i}{\lambda} = 0 \Rightarrow \lambda = \sum_{i=1}^k N_i = N$$

$$\therefore q_i = \frac{N_i}{N} \text{ for each } i \in \{1, \dots, k\}$$

## Univariate Distribution Example : Normal Distribution

Suppose  $N$  sample  $x^{(i)}$  iid

$$x^{(i)} \sim p(x^{(i)}|\theta) = N(x^{(i)}|\mu, \sigma^2) \\ = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}$$

① 第一种方式.

Exponential Family

$$p(x|\theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{1}{2\sigma^2}[x^2 - 2\mu x + \mu^2]\right\} \\ = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{x^2}{2\sigma^2} + \frac{\mu x}{\sigma^2} - \frac{\mu^2}{2\sigma^2}\right\} \\ = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{\mu^2}{2\sigma^2}\right\} \exp\left\{\left[\frac{\mu}{\sigma^2}\right]^T \begin{bmatrix} x \\ x^2 \end{bmatrix}\right\}$$

$$\eta = \begin{bmatrix} \frac{\mu}{\sigma^2} \\ -\frac{1}{2\sigma^2} \end{bmatrix} \quad T(x) = \begin{bmatrix} x \\ x^2 \end{bmatrix}$$

$$\text{then } p(x|\theta) = (2\pi)^{-\frac{1}{2}} (\sigma^2)^{-\frac{1}{2}} \exp\left\{-\frac{\mu^2}{2\sigma^2}\right\} \exp\{\eta^T T(x)\}$$

Calculation

Part

$$-\frac{1}{2\sigma^2} = \eta_2 \Rightarrow \boxed{\sigma^2 = -\frac{1}{2\eta_2}} \Rightarrow \exp\left\{-\frac{1}{2}\log \sigma^2\right\} = \exp\left\{-\frac{1}{2}\log \frac{1}{2\eta_2}\right\}$$

$$\frac{\mu}{\sigma^2} = \eta_1 \Rightarrow \mu = \frac{\eta_1}{2\eta_2} \quad = \exp\left\{\frac{1}{2}\log -2\eta_2\right\}$$

$$\boxed{\mu = \eta_1, \sigma^2 = -\frac{\eta_1}{2\eta_2}}$$

$$\Rightarrow \frac{\mu^2}{\sigma^2} = \frac{\frac{\eta_1^2}{4\eta_2^2}}{-\frac{1}{2\eta_2}} = -\frac{\eta_1^2}{4\eta_2}$$

$$\Rightarrow \frac{-\mu^2}{2\sigma^2} = -\frac{\eta_1^2}{4\eta_2^2} \cdot \left(2 \cdot \left(-\frac{1}{2\eta_2}\right)\right)$$

$$= \frac{\eta_1^2}{4\eta_2}$$

$$\therefore p(x|\theta) = (2\pi)^{-\frac{1}{2}} \exp\{\eta^T T(x) - (-\frac{1}{2}\log -2\eta_2 - \frac{\eta_1^2}{4\eta_2})\}$$

$$\text{and } \eta = \begin{bmatrix} \mu/\sigma^2 \\ -1/2\sigma^2 \end{bmatrix} \quad T(x) = \begin{bmatrix} x \\ x^2 \end{bmatrix} \quad h(x) = (2\pi)^{-\frac{1}{2}} \quad A(\eta) = -\frac{1}{2}\log -2\eta_2 - \frac{\eta_1^2}{4\eta_2}$$

We know the fact that

$$\nabla_{\eta} A(\eta_{MLE}) = \frac{1}{N} \sum_{i=1}^n T(x^{(i)})$$

then  $\nabla_{\eta} \left( \frac{\partial A(\eta)}{\partial \eta_1} \right) = \left( \frac{\frac{1}{N} \sum_{i=1}^n x^{(i)}}{\frac{1}{N} \sum_{i=1}^n (x^{(i)})^2} \right) \quad A(\eta) = -\frac{1}{2}(\log -2\eta_2 - \frac{\eta_1^2}{4\eta_2})$

$$\frac{\partial A(\eta)}{\partial \eta_1} = -\frac{2\eta_1}{4\eta_2} = -\frac{\eta_1}{2\eta_2} = \mu \Rightarrow \hat{\mu}_{MLE} = \frac{1}{N} \sum_{i=1}^n x^{(i)}$$

$$\begin{aligned} \frac{\partial A(\eta)}{\partial \eta_2} &= \frac{-\frac{1}{2}}{-2\eta_2} \cdot (-2) + \frac{\eta_1^2}{4\eta_2^2} = -\frac{1}{2\eta_2} + \left(-\frac{\eta_1}{2\eta_2}\right)^2 \\ &= \sigma^2 + \mu^2 \end{aligned}$$

$$\therefore \hat{\sigma}_{MLE}^2 + \hat{\mu}_{MLE}^2 = \frac{1}{N} \sum_{i=1}^n (x^{(i)})^2$$

$$\begin{aligned} \hat{\sigma}_{MLE}^2 &= \frac{1}{N} \sum_{i=1}^n (x^{(i)})^2 - (\bar{x}_n)^2 \\ &= \frac{1}{N} \sum_{i=1}^n (x^{(i)} - \bar{x}_n)^2 \end{aligned}$$

② 另一种方式 —— MLE 直接求解

$$p(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\Rightarrow \ell(\mu, \sigma^2; D) = (2\pi)^{-\frac{n}{2}} (\sigma^2)^{-\frac{n}{2}} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2}$$

$$\Rightarrow \ell(\mu, \sigma^2; D) = -\frac{n}{2} \log 2\pi - \frac{n}{2} \log (\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

store  $\begin{cases} \frac{\partial \ell}{\partial \mu} = 0 \\ \frac{\partial \ell}{\partial \sigma^2} = 0 \end{cases} \Rightarrow \frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = 0 \Rightarrow \sum_{i=1}^n x_i = n\mu \Rightarrow \hat{\mu}_{MLE} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}_n$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \mu)^2 = 0$$

$$\sigma^2 n = \sum_{i=1}^n (x_i - \mu)^2$$

$$\Rightarrow \hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}_n)^2$$